Panagiotis Fytas

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Cambridge, UK | EU/Pre-Settled UK Status

Summary

PhD researcher at the University of Cambridge specialising in biomedical AI, clinical NLP, multimodal learning and biomedical LLMs. Experienced in developing, validating and evaluating AI/ML models for real-world biomedical problems using EHRs, clinical images, radiology reports and biomedical literature. Equal-contribution first author of a Bioinformatics paper on LLaMA fine-tuning for gene-disease relation extraction, with industry experience building scalable Python/Spark/Kafka systems using Docker, Kubernetes and CI/CD.

Education

University of Cambridge — Cambridge, UK
PhD in Computation, Cognition and Language; Biomedical AI / Clinical NLP — *Sep. 2021 – Present*
Supervised by Prof. Anna Korhonen

Imperial College London — London, UK
MSc in Advanced Computing; Distinction, 85%; Top of the Programme — *Sep. 2020*
Winton Capital Prize for Best MSc Advanced Computing Student

National Technical University of Athens — Athens, Greece
BSc & MEng in Electrical and Computer Engineering; Grade: 8.58/10 — *Jul. 2019*

Experience

PhD Researcher in Biomedical AI — Sep. 2021 – Present
University of Cambridge, Language Technology Lab — *Cambridge, UK*

- Developing and validating multimodal AI models for pathology classification and radiology report generation using chest X-rays, radiology reports and structured Electronic Health Records (EHRs).
- Building deep learning and LLM-based systems for clinical NLP, biomedical information extraction and gene-disease relation extraction.
- Training and evaluating medical imaging and vision-language models for chest X-ray classification, image reconstruction and clinically relevant representation learning.
- Designed robust evaluation pipelines for real-world biomedical data, investigating model interpretability, noisy clinical labels and clinically meaningful performance assessment.
- Ran large-scale model training and evaluation experiments on GPU/HPC infrastructure using Slurm, PyTorch and Hugging Face tooling.

Teaching, Research Supervision and Seminar Organisation — Sep. 2022 – Jul. 2025
University of Cambridge — *Cambridge, UK*

- Supervised students in the Computational Linguistics module.
- Co-supervised a year-long Master's research project.
- Organised the Language Technology Lab research seminar series, inviting speakers, coordinating logistics and chairing research discussions across NLP and machine learning.

Software Engineer — Jan. 2019 – Aug. 2019
Performance Technologies S.A. — *Athens, Greece*

- Developed scalable Python/PySpark Structured Streaming pipelines for the largest telecommunications provider in Greece.
- Built fault-tolerant database replication workflows from Oracle databases through Apache Kafka into Vertica.
- Used Docker, Git, Jenkins CI/CD and HDFS infrastructure to support reproducible, production-oriented development.

Selected Research Projects

Rethinking Clinical Relevance in Chest X-ray Machine Learning | *ML evaluation, Data Quality* — 2026

- Systematically evaluated how reference choices affect performance estimates and model rankings in CXR pathology classification and image quality assessment.
- Compared supervised image classifiers, self-supervised models and fine-tuned vision-language models, including MedKLIP, GLoRIA and ConVIRT.

- Collected paired expert image and report labels for thoracic findings from a Cambridge University Hospitals cohort and curated a subset of MIMIC-CXR for controlled comparison.
- Showed that commonly used report-derived labels and generic IQA metrics such as SSIM and PSNR can diverge from expert clinical judgement.

Multimodal Clinical Representation Learning | *VLMs, EHR, CXR, radiology reports* — In progress

- Fine-tuning vision-language models to integrate multimodal EHR context with chest X-ray images and radiology reports.
- Learning clinically enriched image embeddings from imaging, text and structured clinical measurements for downstream medical AI tasks.
- Evaluating representations on pathology classification, radiology report generation and transfer to clinically relevant prediction tasks.

BioTriplex | *LLaMA 3.1, biomedical NLP, relation extraction, full-text annotation* — 2025

- Built a full-text biomedical corpus of 100 research articles annotated for disease names, genes and 21 disease-gene relation subtypes.
- Fine-tuned LLaMA 3.1 8B for gene-disease relation extraction, outperforming zero-/few-shot GPT-4, Claude Sonnet 3.7 and LLaMA baselines.
- Code and data at github.com/PanagiotisFytas/BioTriplex.

RadPrompt | *LLMs, radiology reports, clinical NLP, rule-based prompting* — 2024

- Developed an LLM-oriented methodology for radiology report classification using rule-based clinical insights.
- State-of-the-art performance for radiology report classification, boosting LLM performance with rule-based expert systems.

Parallelizing DPOR Algorithms in Concuerror | *Erlang, Bash, distributed systems* — 2018 – 2019

- Parallelised dynamic partial order reduction algorithms for testing and verifying concurrent Erlang programs.
- Implemented distributed Erlang/OTP systems that scaled verification workloads to hundreds of workers.

Selected Publications

BioTriplex: a full-text annotated corpus for fine-tuning language models in gene-disease relation extraction tasks — 2026

- *Bioinformatics*. Equal-contribution first author. Introduced a full-text biomedical corpus for gene–disease relation extraction; fine-tuned LLMs outperform zero-/few-shot commercial LLM baselines.

Rethinking Clinical Relevance in Chest X-ray Machine Learning: How Evaluation References Define Performance — Under review

- *Submitted to Medical Image Analysis*. Evaluates how reference labels and IQA metrics affect CXR model performance estimates, model rankings and clinical validity.

Can Rule-Based Insights Enhance LLMs for Radiology Report Classification? Introducing the RadPrompt Methodology — 2024

- *BioNLP / ACL*. Developed an LLM-oriented methodology for radiology report classification using rule-based clinical insights to boost LLM performance.

What Makes a Scientific Paper be Accepted for Publication? — 2021

- *Causal Inference / EMNLP*. Combined NLP, deep learning, causal inference and explainability for peer-review text analysis.

Awards

Winton Capital Prize for Best MSc Advanced Computing Student, Imperial College London — 2020
Harding Distinguished Postgraduate Scholarship, University of Cambridge — 2021

Technical Skills

Languages: Python, Bash, Java, C/C++, Erlang, SQL, LaTeX
ML / AI: PyTorch, Hugging Face Transformers/Datasets, vLLM, LLaMA-Recipes, scikit-learn, pandas, NumPy
LLM Systems: RAG, tool use, structured outputs, prompt engineering, LLM evaluation, Weights & Biases, LangChain
Biomedical AI: Clinical NLP, chest X-rays, clinical imaging, clinical labels, biomedical relation extraction, EHRs
Data Engineering: Apache Spark Structured Streaming, Apache Kafka, HDFS, Vertica, PostgreSQL
ML Engineering: FastAPI, Kubernetes, Docker, Git, Jenkins CI/CD, unit testing, Slurm, Linux, Google Cloud Platform